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What is a Vector Database?
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=> A vector database stores data in the form of numbers called vectors/embeddings.

=> These vectors represent the meaning of text.

Example Text : Python is used for AI development.

This text is converted into numbers like below:

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[0.23, -0.18, 0.91, 0.44, ...]
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These numbers are called embeddings.

=> The vector database stores these embeddings and helps search based on meaning, not only exact keywords.

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What is an Embedding?
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=> An embedding is a numerical representation of text.

Example: "I want to learn Python"

and

"How can I study Python?"

=> These two sentences have different words, but the meaning is similar.

=> A normal database may not understand this similarity.

=> A vector database understands because both sentences will have similar vector values.

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Popular Vector Databases
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1) Chroma DB : Best for beginners and local projects

2) PineCone : Cloud vector database

3) FAISS : Fast local vector search

4) Weaviate : Cloud/native vector database

5) Milvus : Large-scale vector database

6) Qdrant : Open-source vector database

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What is RAG?

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** RAG stands for Retrieval-Augmented Generation.

** It is a technique used in Generative AI where an LLM does not depend only on its own trained knowledge. Instead, it first retrieves relevant information from our own data and then generates an answer using that information.

** RAG means giving external knowledge to an AI model before asking it to answer.

Suppose we ask ChatGPT:

What is the refund policy of Ashok IT?

If the model was not trained on Ashok IT latest policy document, it may give a wrong or generic answer.

With RAG:

We upload Ashok IT policy documents.
The system searches the relevant policy content.
The LLM receives that content.
The LLM answers based on that content.

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Why Do We Need RAG?

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=> LLMs have some limitations.

Problem 1: LLM does not know private data

Problem 2: LLM knowledge may be outdated (Training cutoff)

Problem 3: LLM can hallucinate

Note : RAG reduces all the above problems, because the model answers using provided documents.

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RAG Full Form Breakdown

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R : Retrieval : The system searches relevant information from documents.

A : Augmented : The retrieved information is added to the user question.

G : Generation : The LLM generates the final answer.